Q8) O potencial elétrico criado por uma esfera carregada é o potencial na superfície desta esfera. Assim,

$$V = k\frac{Q}{R}, \quad \text{onde} \quad k = \frac{1}{4\pi\varepsilon_0} = 8{,}99\times10^{9}\ N\frac{m^2}{C^2}$$

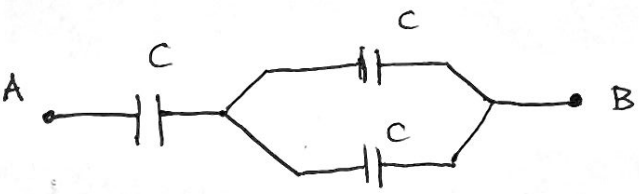
$$R = 3{,}0\times10^{-7}\ m.$$

Assim,

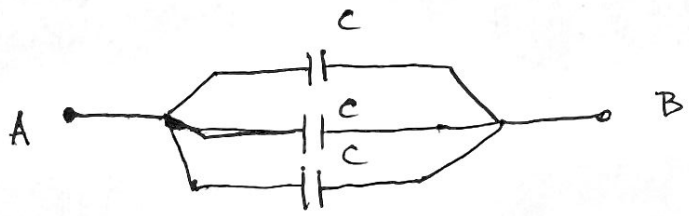
$$Q = \frac{VR}{k} = \frac{-(0{,}15\,V)\times(3{,}0\times10^{-7}\,m)}{8{,}99\times10^{9}\ N\frac{m^2}{C^2}} \Longrightarrow$$

$$Q = -0{,}50\times10^{-16}\ C \Rightarrow Q = -5{,}0\times10^{-18}\ C\ //.$$

Q9)

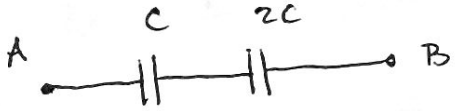


(1)



(2)

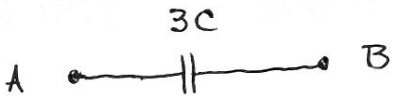
O capacitor ①:



$$(1) \Rightarrow \frac{1}{C_1} = \frac{1}{C} + \frac{1}{2C} = \frac{3}{2C}$$

$$\Rightarrow C_1 = \frac{2}{3}C\ //$$

O capacitor ②:



$$\Rightarrow C_2 = 3C$$

Então, $\frac{C_1}{C_2} = \frac{2}{3}C\cdot\frac{1}{3C} = \frac{2}{9}\ //$

Q15) O campo é nulo dentro de um condutor.